

LESSON 2: PRESSURE & ITS MEASUREMENT

Pressure of a Liquid

Consider a fluid in a vessel which exerts force at all points on the sides including the bottom of the container.

If, $P =$ The force and

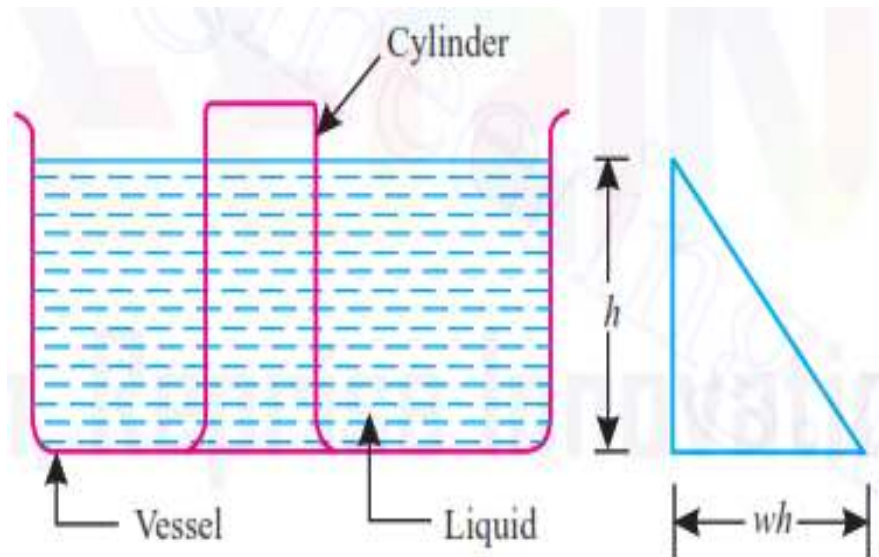
$A =$ Area on which the force acts

Then, Intensity of pressure, $p = \frac{P}{A}$

Pressure Head of a Liquid

Pressure by a fluid can be generated from the concept that fluids generate pressure due to its own weight and this pressure increases as the depth of the liquid increases.

Consider a vessel containing liquid in the fig. below, with the liquid exerting pressure on all sides and the bottom of the container.



Let, h = Height of liquid in the cylinder,
 A = Area of the cylinder base,
 w = Specific weight of the liquid,
and, p = Intensity of pressure.

Now, Total pressure on the base of the cylinder = Weight of liquid in the cylinder

i.e.,

$$p \cdot A = wAh$$

$$p = \frac{wAh}{A} = wh$$

$$\text{i.e., } p = wh$$

As $p = wh$, the intensity of pressure at any point in a liquid depends on the height of the free surface above the point. From the expression,

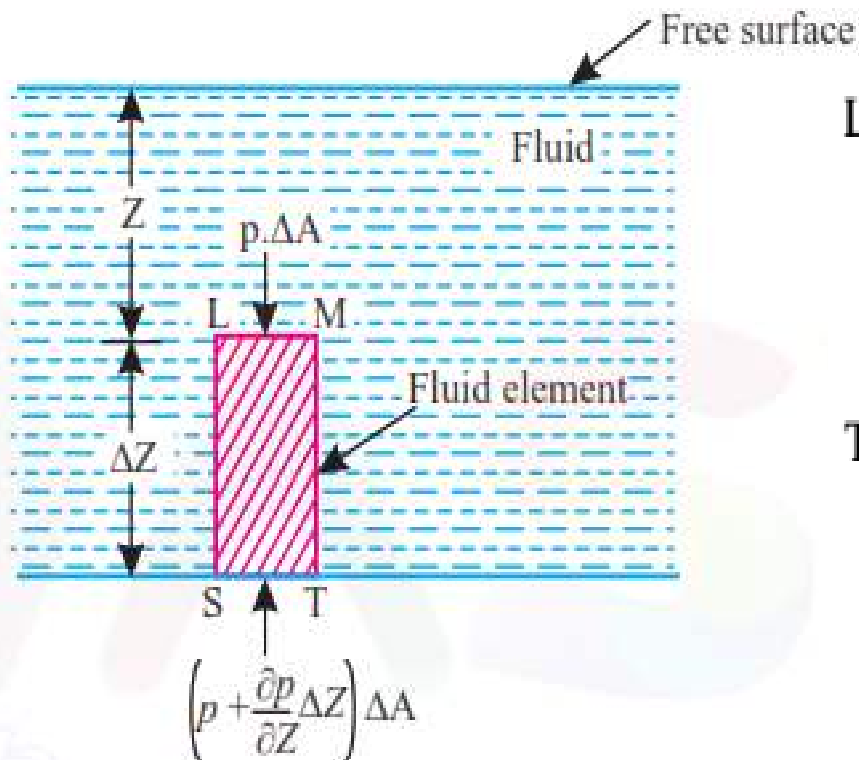
$$h = \frac{p}{w}$$

Pressure Variation in Fluid at Rest:

To determine the pressure at any point in a fluid at rest, “**hydrostatic law**” is used. The law states that:

“The rate of increase of pressure in a vertically downward direction must be equal to the specific weight of the fluid at that point.”

To prove the principle, consider a small fluid element as shown in the fig below;



Let ΔA = Cross-sectional area of element

ΔZ = Height of fluid element

p = Pressure on face AB

Z = Distance of fluid element from free surface.

The forces acting on the fluid element are :

1. Pressure force on $AB = p \times \Delta A$ and acting perpendicular to face AB in the downward direction.

2. Pressure force on $CD = \left(p + \frac{\partial p}{\partial Z} \Delta Z \right) \times \Delta A$, acting perpendicular to face CD , vertically upward

direction.

3. Weight of fluid element = Density $\times g \times$ Volume = $\rho \times g \times (\Delta A \times \Delta Z)$.

4. Pressure forces on surfaces BC and AD are equal and opposite. For equilibrium of fluid element, we have

$$p\Delta A - \left(p + \frac{\partial p}{\partial Z} \Delta Z \right) \Delta A + \rho \times g \times (\Delta A \times \Delta Z) = 0$$

$$p\Delta A - p\Delta A - \frac{\partial p}{\partial Z} \Delta Z \Delta A + \rho \times g \times \Delta A \times \Delta Z = 0$$

$$- \frac{\partial p}{\partial Z} \Delta Z \Delta A + \rho \times g \times \Delta A \Delta Z = 0$$

$$\frac{\partial p}{\partial Z} \Delta Z \Delta A = \rho \times g \times \Delta A \Delta Z \quad \text{or} \quad \frac{\partial p}{\partial Z} = \rho \times g \quad [\text{cancelling } \Delta A \Delta Z \text{ on both sides}]$$

$$\therefore \quad \frac{\partial p}{\partial Z} = \rho \times g = w \quad (\because \rho \times g = w)$$

where w = Weight density of fluid.

Integrating the equation above,

$$\int dp = \int \rho g dZ$$
$$p = \rho g Z$$

Where p is the pressure above atmospheric pressure and Z is the height of the point from free surfaces.

Thus,

$$Z = \frac{p}{\rho \times g}$$

Z is called the **pressure head**

Worked Examples:

1. The pressure at a point in a fluid is given as 3.924 N/cm^2 . Find the corresponding height of fluid when the fluid is: a) water, b) oil of sp gr 0.9.

Solution. Given :

Pressure intensity,
$$p = 3.924 \frac{\text{N}}{\text{cm}^2} = 3.924 \times 10^4 \frac{\text{N}}{\text{m}^2}.$$

The corresponding height, Z , of the fluid is given by equation (2.6) as

$$Z = \frac{P}{\rho \times g}$$

(a) For water,
$$\rho = 1000 \text{ kg/m}^3$$

$\therefore$
$$Z = \frac{p}{\rho \times g} = \frac{3.924 \times 10^4}{1000 \times 9.81} = \mathbf{4 \text{ m of water. Ans.}}$$

(b) For oil, sp. gr.
$$= 0.9$$

$\therefore$ Density of oil
$$\rho_0 = 0.9 \times 1000 = 900 \text{ kg/m}^3$$

$\therefore$
$$Z = \frac{p}{\rho_0 \times g} = \frac{3.924 \times 10^4}{900 \times 9.81} = \mathbf{4.44 \text{ m of oil. Ans.}}$$

Worked Examples:

2. An open tank contains water up to a depth of 2 m and above it an oil of sp. gr of 0.9 for a depth of 1 m. Find the pressure intensity (i) at the interface of the two liquids, and (ii) at the bottom of the tank.

Solution. Given :

Height of water,

$$Z_1 = 2 \text{ m}$$

Height of oil,

$$Z_2 = 1 \text{ m}$$

Sp. gr. of oil,

$$S_0 = 0.9$$

Density of water,

$$\rho_1 = 1000 \text{ kg/m}^3$$

Density of oil,

$$\begin{aligned}\rho_2 &= \text{Sp. gr. of oil} \times \text{Density of water} \\ &= 0.9 \times 1000 = 900 \text{ kg/m}^3\end{aligned}$$

Pressure intensity at any point is given by

$$p = \rho \times g \times Z.$$

(i) At interface, i.e., at A

$$\begin{aligned}p &= \rho_2 \times g \times 1.0 \\ &= 900 \times 9.81 \times 1.0 \\ &= 8829 \frac{\text{N}}{\text{m}^2} = \frac{8829}{10^4} = \mathbf{0.8829 \text{ N/cm}^2. \text{ Ans.}}\end{aligned}$$

(ii) At the bottom, i.e., at B

$$\begin{aligned}p &= \rho_2 \times g Z_2 + \rho_1 \times g \times Z_1 = 900 \times 9.81 \times 1.0 + 1000 \times 9.81 \times 2.0 \\ &= 8829 + 19620 = 28449 \text{ N/m}^2 = \frac{28449}{10^4} \text{ N/cm}^2 = \mathbf{2.8449 \text{ N/cm}^2. \text{ Ans.}}\end{aligned}$$

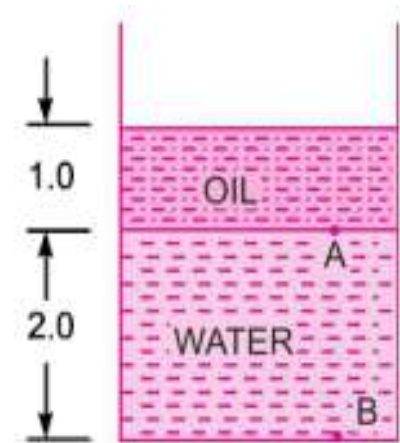


Fig. 2.4

Pressure Measurement

Pressure in a fluid is measured by the following devices:

i) Manometers ii) Mechanical gauges

1. Manometers: Are defined as devices used for measuring the pressure at a point in a fluid by balancing the column of fluid by the same or another column of the fluid.

Manometers are classified as:

(a) Simple manometers: (i) Piezometer, (ii) U-tube manometer, and (iii) Single column manometer.

(b) Differential manometers.

2. Mechanical gauges: These are the devices in which the pressure is measured by balancing the fluid column by spring (elastic element) or dead weight.

Generally, these gauges are used for measuring high pressure and where high precision is not required. Some commonly used mechanical gauges are:

(i) Bourdon tube pressure gauge, (ii) Diaphragm pressure gauge,
(ii) (iii) Bellow pressure gauge, and (iv) Dead-weight pressure gauge

Manometers

Simple Manometers

A “simple manometer” is one which consists of a glass tube whose one end is connected to a point where pressure is to be measured and the other end remains open to atmosphere.

Common Types of Simple Manometers

1. Piezometer:

A piezometer is the simplest form of manometer which can be used for measuring ***moderate pressures*** of liquids. One end of the manometer is connected to the point where pressure is to be measured and the other end is open to the atmosphere as shown as shown in Fig 2.8.

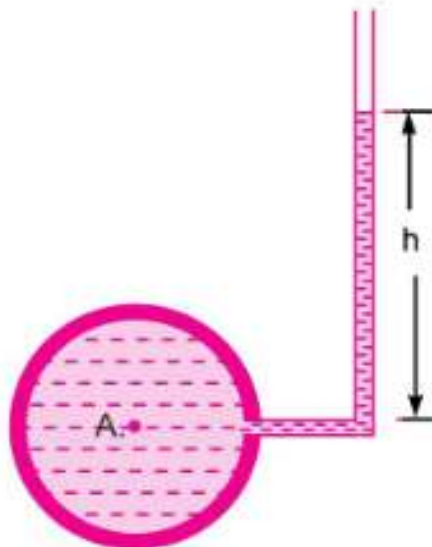


Fig. 2.8 Piezometer.

The rise of the liquid gives the pressure head at that point. If at a point A, the height of liquid say is h in piezometer tube, then pressure at A is given as:

$$= \rho \times g \times h \frac{\text{N}}{\text{m}^2}.$$

Manometers

2. U-tube Manometers:

Consists of glass tube bent in U-shape, one end of which is connected to a point at which pressure is to be measured and the other end remains open to the atmosphere as shown in fig 2.9

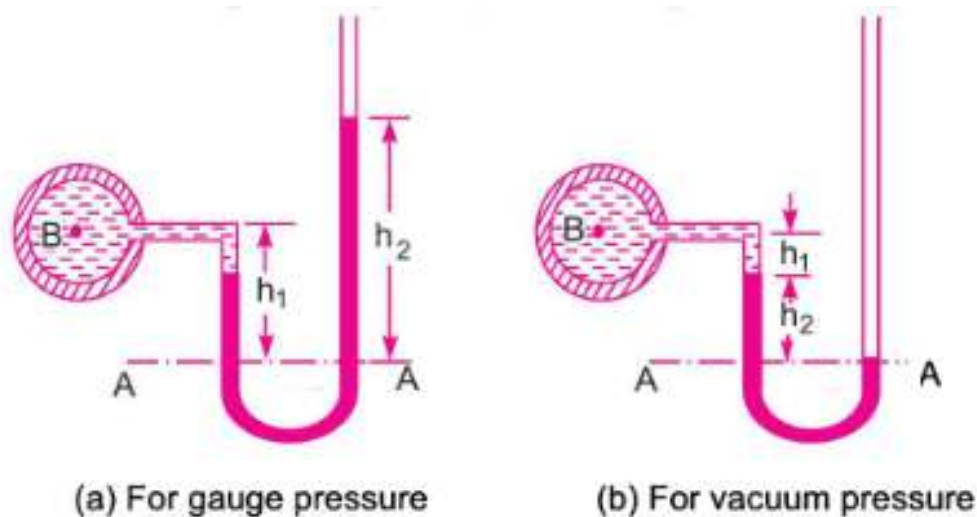


Fig. 2.9 U-tube Manometer.

The rise of the liquid gives the pressure head at that point. If at a point A, the height of liquid say is h in piezometer tube, then pressure at A is given as:

a) For Gauge Pressure, let B be the point at which pressure is to be measured, whose value is p . The datum line is A-A

Let

- h_1 = Height of light liquid above the datum line
- h_2 = Height of heavy liquid above the datum line
- S_1 = Sp. gr. of light liquid
- ρ_1 = Density of light liquid = $1000 \times S_1$
- S_2 = Sp. gr. of heavy liquid
- ρ_2 = Density of heavy liquid = $1000 \times S_2$

Manometers

Pressure at horizontal surface is the same, hence pressure above horizontal datum A-A (Right side and left side should be the same)

$$\text{Pressure above A-A in the left column} = p + \rho_1 \times g \times h_1$$

$$\text{Pressure above A-A in the right column} = \rho_2 \times g \times h_2$$

$$\text{Hence equating the two pressures} \quad p + \rho_1 g h_1 = \rho_2 g h_2$$

$$\therefore \quad p = (\rho_2 g h_2 - \rho_1 \times g \times h_1).$$

a) For Vacuum Pressure, To measure the vacuum pressure, consider the level of heavy liquid in the manometer (Fig 2b)

$$\text{Pressure above A-A in the left column} = \rho_2 g h_2 + \rho_1 g h_1 + p$$

Pressure head in the right column above A-A = 0

$$\rho_2 g h_2 + \rho_1 g h_1 + p = 0$$

$$p = -(\rho_2 g h_2 + \rho_1 g h_1).$$

Worked Examples

1. A U-tube manometer is used to measure the pressure of oil of sp gr of 0.85 flowing in a pipeline. Its left end is connected to the pipe and the right- limb is open to the atmosphere. The centre of the pipe is 100 mm below the level of mercury (sp gr =13.6) in the right limb. If the difference of mercury level in the two limbs is 160 mm, determine the absolute pressure of the oil in the pipe.

Solution. Specific gravity of oil, $S_1 = 0.85$

Specific gravity of mercury, $S_2 = 13.6$

Height of the oil in the left limb,

$$h_1 = 160 - 100 = 60 \text{ mm} = 0.06 \text{ m}$$

Difference of mercury level,

$$h_2 = 160 \text{ mm} = 0.16 \text{ m.}$$

Absolute pressure of oil:

Let, h_1 = Gauge pressure in the pipe in terms of head of water, and

p = Gauge pressure in terms of kN/m^2 .

Equating the pressure heads above the datum line $X-X$, we get:

$$h + h_1 S_1 = h_2 S_2$$

$$\text{or, } h + 0.06 \times 0.85 = 0.16 \times 13.6 = 2.125 \text{ m}$$

The pressure p is given by:

$$p = wh$$

$$= 9.81 \times 2.125 \text{ kN/m}^2$$

$$= 20.84 \text{ kPa} \quad (\because w = 9.81 \text{ kN/m}^3 \text{ in S.I. units})$$

Absolute pressure of oil in the tube,

$$P_{abs.} = P_{atm.} + P_{gauge}$$

$$= 100 + 20.84 = \mathbf{120.84 \text{ kPa (Ans.)}}$$

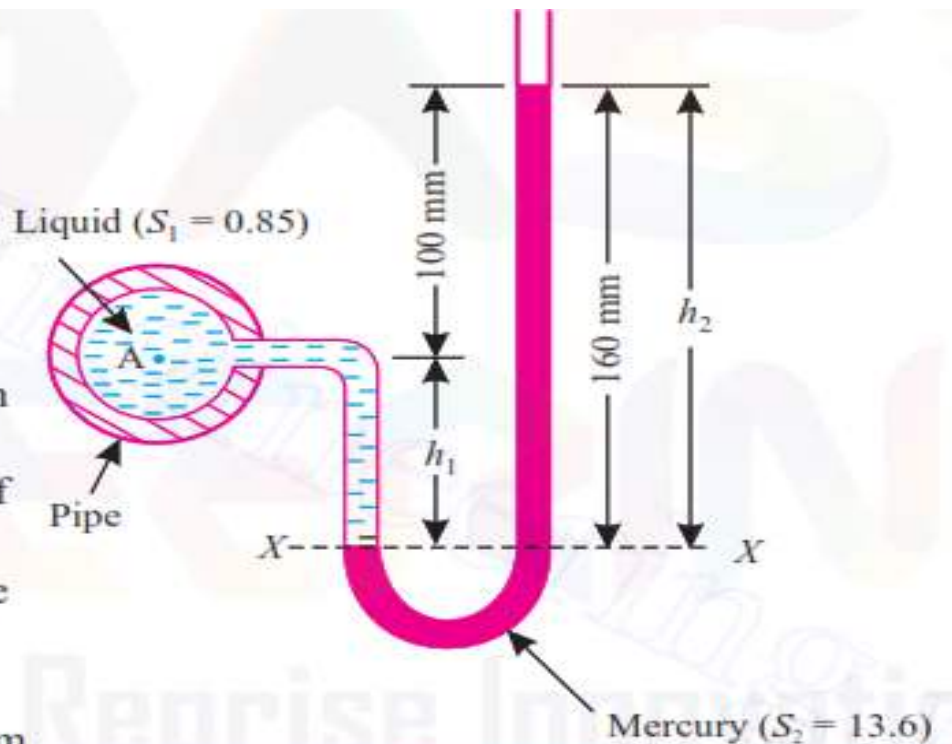


Fig. 2.13

Worked Examples

2. Fig. 2.12 shows a conical vessel having its outlet at A to which a U-tube manometer is connected. The reading of the manometer given in the figure shows when the vessel is empty. Find the reading of the manometer when the vessel is completely filled with water.

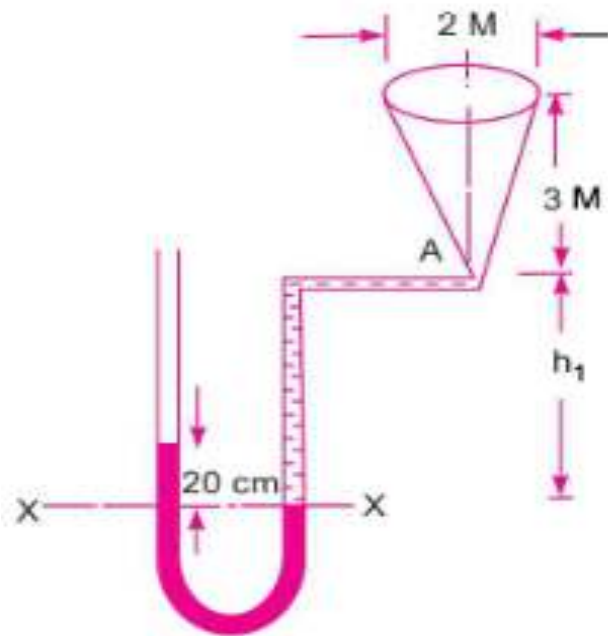


Fig. 2.12

Solution. Vessel is empty. Given :

Difference of mercury level $h_2 = 20 \text{ cm}$

Let $h_1 =$ Height of water above X-X

Sp. gr. of mercury, $S_2 = 13.6$

Sp. gr. of water, $S_1 = 1.0$

Density of mercury, $\rho_2 = 13.6 \times 1000$

Density of water, $\rho_1 = 1000$

Equating the pressure above datum line X-X, we have

$$\rho_2 \times g \times h_2 = \rho_1 \times g \times h_1$$

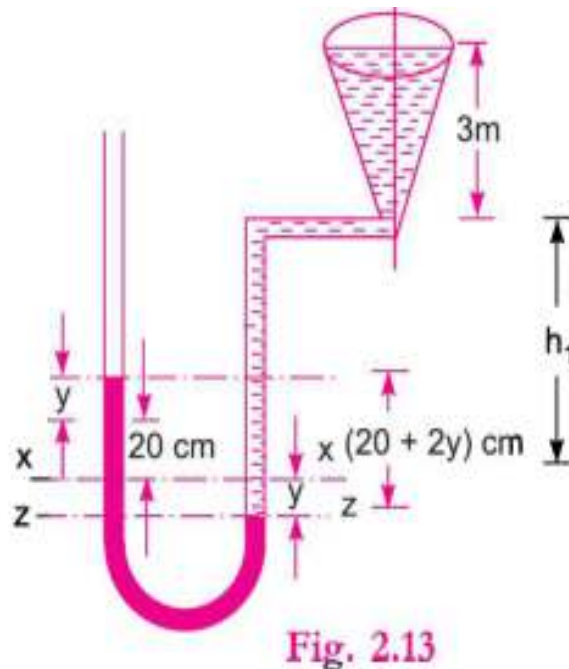
$$\text{or } 13.6 \times 1000 \times 9.81 \times 0.2 = 1000 \times 9.81 \times h_1$$

$$h_1 = 2.72 \text{ m of water.}$$

When the vessel is full of water.

When the vessel is full of water, the pressure in the right limb will increase and the mercury level in the right limb will go down. Let the distance through which mercury goes down in the right limb be $y \text{ cm}$ as shown in the Fig 2.13. The mercury will rise in the left by a distance of $y \text{ cm}$. Now the datum line Z-Z. Equating the pressure above the datum line Z-Z.

Worked Examples



Pressure in left limb = Pressure in right limb

$$13.6 \times 1000 \times 9.81 \times (0.2 + 2y/100)$$

$$= 1000 \times 9.81 \times (3 + h_1 + y/100)$$

or $13.6 \times (0.2 + 2y/100) = (3 + 2.72 + y/100) \quad (\because h_1 = 2.72 \text{ cm})$

or $2.72 + 27.2y/100 = 3 + 2.72 + y/100$

or $(27.2y - y)/100 = 3.0$

or $26.2y = 3 \times 100 = 300$

$$\therefore y = \frac{300}{26.2} = 11.45 \text{ cm}$$

The difference of mercury level in two limbs

$$= (20 + 2y) \text{ cm of mercury}$$

$$= 20 + 2 \times 11.45 = 20 + 22.90$$

$$= 42.90 \text{ cm of mercury}$$

$\therefore$ Reading of manometer = **42.90 cm. Ans.**

Manometers

3. Single Column Manometers (Micro-Manometers):

Single column manometers are modified form of a U-tube manometers in which a reservoir, having a large cross-sectional area (100 times) as compared to the area of the tube is connected to one of the limbs, (say left limb) as shown in fig. 2.15.

There are two types of single column manometers:

- i) *Vertical single column manometer* ii) *Inclined single column manometer.*

i) Vertical Single Column Manometer

Consider the arrangement in Fig. 2.15 as single column manometer, with X-X as the datum line in the reservoir. When the manometer is connected to the pipe, the high pressure at A pushes the heavy liquid in the reservoir downward to rise in the right limb.

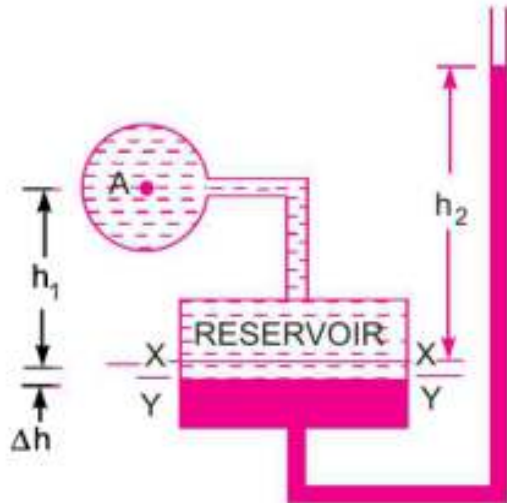


Fig. 2.15 Vertical single column manometer.

Let Δh = Fall of heavy liquid in reservoir

h_2 = Rise of heavy liquid in right limb

h_1 = Height of centre of pipe above X-X

p_A = Pressure at A, which is to be measured

A = Cross-sectional area of the reservoir

a = Cross-sectional area of the right limb

S_1 = Sp. gr. of liquid in pipe

S_2 = Sp. gr. of heavy liquid in reservoir and right limb

ρ_1 = Density of liquid in pipe

ρ_2 = Density of liquid in reservoir

Manometers

A fall of heavy liquid in the reservoir will lead to a rise in heavy liquid in the limb

$$A \times \Delta h = a \times h_2$$

$$\Delta h = \frac{a \times h_2}{A}$$

Now consider the datum line Y-Y as shown in Fig 2.15. Then pressure in the right limb above Y-Y is given as:

$$= \rho_2 \times g \times (\Delta h + h_2)$$

$$\text{Pressure in the left limb above Y-Y} = \rho_1 \times g \times (\Delta h + h_1) + p_A$$

Equating these pressures, we have

$$\rho_2 \times g \times (\Delta h + h_2) = \rho_1 \times g \times (\Delta h + h_1) + p_A$$

or

$$p_A = \rho_2 g (\Delta h + h_2) - \rho_1 g (\Delta h + h_1)$$

$$= \Delta h [\rho_2 g - \rho_1 g] + h_2 \rho_2 g - h_1 \rho_1 g$$

$$\text{But from equation (i),} \quad \Delta h = \frac{a \times h_2}{A}$$

$$\therefore p_A = \frac{a \times h_2}{A} [\rho_2 g - \rho_1 g] + h_2 \rho_2 g - h_1 \rho_1 g$$

Since area A is very large as compared to a, ratio a/A becomes very small and can be neglected, hence:

$$\text{Then } p_A = h_2 \rho_2 g - h_1 \rho_1 g$$

Manometers

ii). Inclined Single Column Manometers

Consider an inclined single column manometer. The manometer is more sensitive and due to inclination, the distance moved by the heavy liquid in the right limb will be more.

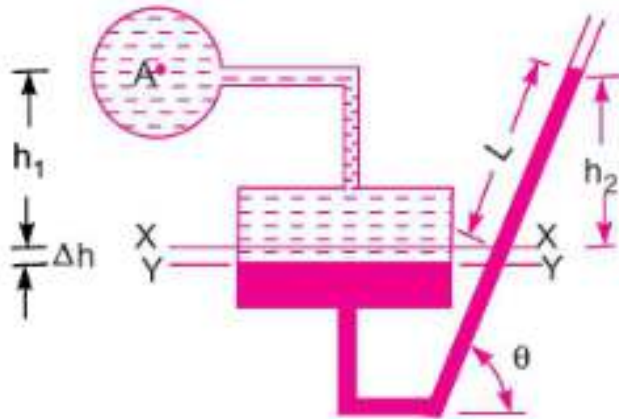


Fig. 2.16 Inclined single column manometer.

Let L = Length of heavy liquid moved in right limb from X-X

θ = Inclination of right limb with horizontal

h_2 = Vertical rise of heavy liquid in right limb from X-X = $L \times \sin \theta$

From equation (2.10), the pressure at A is

$$p_A = h_2 \rho_2 g - h_1 \rho_1 g.$$

Substituting the value of h_2 , we get

$$p_A = \sin \theta \times \rho_2 g - h_1 \rho_1 g.$$

Manometers

Worked Example

A single column manometer is connected to a pipe containing a liquid of sp gr 0.9 as shown in Fig. 2.17. Find the pressure in the pipe if the area of the reservoir is 100 times the area of the tube for manometer reading shown in the Fig. 2.17. Take the sp gr of mercury as 13.6.

Solution. Given :

Sp. gr. of liquid in pipe, $S_1 = 0.9$
 $\therefore$ Density $\rho_1 = 900 \text{ kg/m}^3$
 Sp. gr. of heavy liquid, $S_2 = 13.6$
 Density, $\rho_2 = 13.6 \times 1000$

$$\frac{\text{Area of reservoir}}{\text{Area of right limb}} = \frac{A}{a} = 100$$

Height of liquid, $h_1 = 20 \text{ cm} = 0.2 \text{ m}$

Rise of mercury in right limb,
 $h_2 = 40 \text{ cm} = 0.4 \text{ m}$

Let $p_A = \text{Pressure in pipe}$

Using equation (2.9), we get

$$\begin{aligned} p_A &= \frac{a}{A} h_2[\rho_2 g - \rho_1 g] + h_2 \rho_2 g - h_1 \rho_1 g \\ &= \frac{1}{100} \times 0.4[13.6 \times 1000 \times 9.81 - 900 \times 9.81] + 0.4 \times 13.6 \times 1000 \times 9.81 - 0.2 \times 900 \times 9.81 \\ &= \frac{0.4}{100} [133416 - 8829] + 53366.4 - 1765.8 \\ &= 533.664 + 53366.4 - 1765.8 \text{ N/m}^2 = 52134 \text{ N/m}^2 = 5.21 \text{ N/cm}^2. \text{ Ans.} \end{aligned}$$

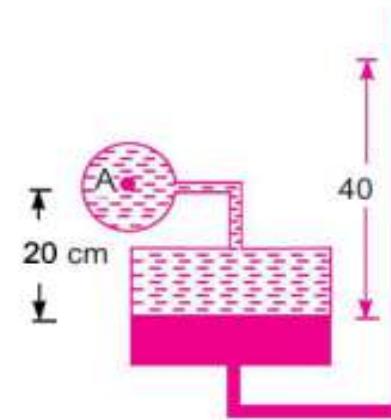


Fig. 2.17

Manometers

Differential Manometers

A differential manometer is used to measure the difference in pressures between two points in a pipe, or in two different pipes. In its simplest form a differential manometer consists of a U-tube, containing a heavy liquid, whose two ends are connected to the points, whose difference of pressures is required to be found out.

The following are the most commonly used types of differential manometers:

1. U-tube differential manometer.
2. Inverted U-tube differential manometer.

1. U-tube differential manometer: A U-tube differential manometer is shown in Fig. 2.21 (a) shows a differential manometer whose two ends are connected with two different points A and B at the same level and containing same liquid. Let,

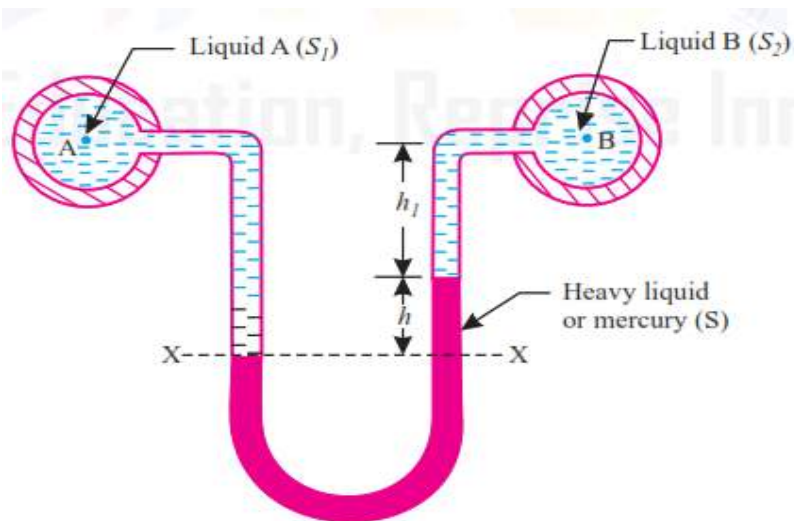


Fig. 2.21. (a) Two pipes at same level.

- h = Difference of mercury levels (heavy liquid) in the U-tube,
- h_1 = Distance of the centre of A from the mercury level in the right limb,
- $S_1 (= S_2)$ = Specific gravity of liquid at the two points A and B
- S = Specific gravity of heavy liquid or mercury in the U-tube,
- h_A = Pressure head at A, and
- h_B = Pressure head at B.

Manometers

We know that the pressures in the left limb and right limb, above the datum line are equal.

Pressure head in the left limb $= h_A + (h_1 + h) S_1$

Pressure head in the *right limb*

$$= h_B + h_1 \times S_1 + h \times S$$

$$h_A + (h_1 + h)S_1 = h_B + h_1S_1 + hS$$

or,
$$h_A - h_B = h_1S_1 + hS - (h_1 + h) S_1$$
$$= h_1 S_1 + hS - h_1S_1 - hS_1 = h (S - S_1)$$

i.e., Difference of pressure head,

$$h_A - h_B = h (S - S_1)$$

Manometers

Worked Example:

A differential manometer is connected at the two points A and B as shown in Fig 2.20. At B, air pressure is 9.81 N/cm^2 (abs), find the absolute pressure at A.

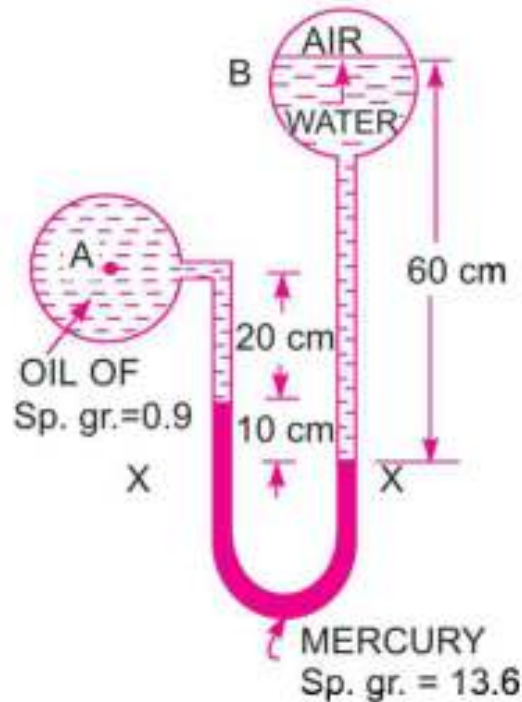


Fig. 2.20

Solution. Given :

Air pressure at

$$B = 9.81 \text{ N/cm}^2$$

or

$$p_B = 9.81 \times 10^4 \text{ N/m}^2$$

Density of oil

$$= 0.9 \times 1000 = 900 \text{ kg/m}^3$$

Density of mercury

$$= 13.6 \times 1000 \text{ kg/m}^3$$

Let the pressure at A is p_A

Taking datum line at X-X

Pressure above X-X in the right limb

$$\begin{aligned} &= 1000 \times 9.81 \times 0.6 + p_B \\ &= 5886 + 98100 = 103986 \end{aligned}$$

Pressure above X-X in the left limb

$$\begin{aligned} &= 13.6 \times 1000 \times 9.81 \times 0.1 + 900 \\ &\quad \times 9.81 \times 0.2 + p_A \\ &= 13341.6 + 1765.8 + p_A \end{aligned}$$

Equating the two pressure heads

$$103986 = 13341.6 + 1765.8 + p_A$$

$\therefore$

$$p_A = 103986 - 15107.4 = 88876.8$$

$\therefore$

$$p_A = 88876.8 \text{ N/m}^2 = \frac{88876.8 \text{ N}}{10000 \text{ cm}^2} = 8.887 \frac{\text{N}}{\text{cm}^2}$$

$\therefore$ Absolute pressure at A = 8.887 N/cm^2 . Ans.

Manometers

Inverted U-tube differential manometers

This type of manometer is used to measure difference of two pressures where accuracy is the major consideration.

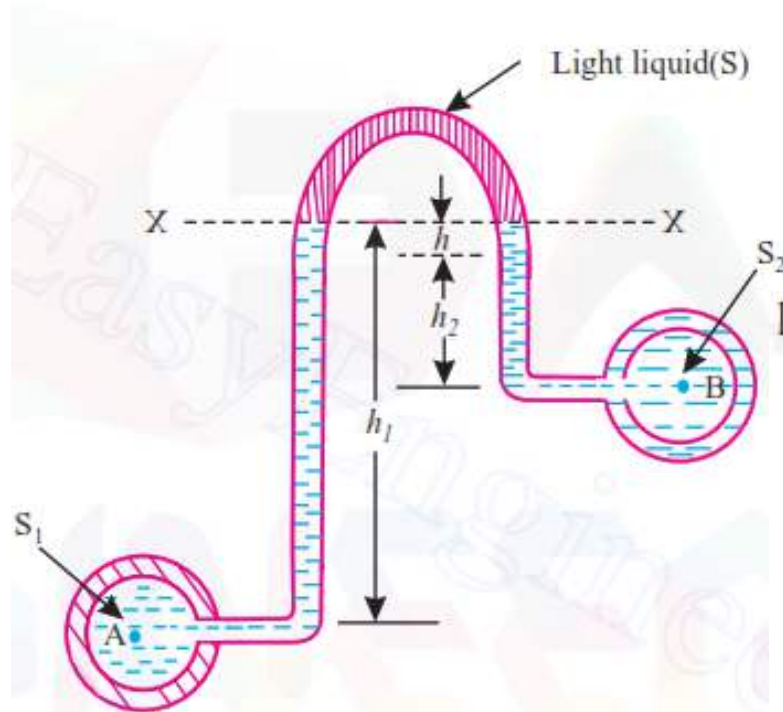


Fig. 2.24. Inverted differential manometer.

Fig. 2.24 illustrates an inverted U-tube, containing light liquid, whose two ends are connected to the two points (A&B) whose difference in pressures is to be determined.

Let pressure at A, be more than at B

Let, h_1 = Height of liquid in the left limb below the datum line X-X,
 h_2 = Height of liquid in the right limb below the datum line,
 h = Difference of levels of the light liquid in the right and left limbs (also known as manometer reading),

S_1 = Specific gravity of the liquid in the left limb,

S_2 = Specific gravity of the liquid in the right limb,

S = Specific gravity of the light liquid,

h_A = Pressure head at A, and

h_B = Pressure head at B.

...and we know that pressure heads at right limb is equal to that in the left below X-X

Pressure head in the *left limb* below X-X = $h_A - h_1 \times S_1$

Pressure head in the *right limb* below X-X = $h_B - h_2 \times S_2 - h \times S$

Equating the above heads, we get:

$$h_A - h_1 \times S_1 = h_B - h_2 \times S_2 - h \times S$$

$$h_A - h_B = h_1 \times S_1 - h_2 \times S_2 - h \times S$$

$$h_A - h_B = h_1 S_1 - h_2 S_2 - h S$$

Worked Examples

An inverted differential manometer is connected to two pipes A & B carrying water under pressure as shown in Fig. 2.26. The fluid in the manometer is oil of Sp gr of 0.75. Determine the pressure difference between A & B.

Solution. Specific gravity of oil, $S = 0.75$

Specific gravity of water, $S_1, S_2 = 1$

Difference of oil in the two limbs = $(450 + 200) - 450 = 200$ mm

We know that pressure heads on the left and right limbs below the datum line $X-X$ are equal.

Pressure head in the left limb below $X-X$

$$= h_A - \frac{450}{1000} \times 1 = h_A - 0.45$$

Pressure head in the right limb below $X-X$

$$\begin{aligned} &= h_B - \frac{450}{1000} \times 1 - \frac{200}{1000} \times 0.75 \\ &= h_B - 0.45 - 0.15 = h_B - 0.6 \end{aligned}$$

Equating the two pressure heads, we get:

$$h_A - 0.45 = h_B - 0.6$$

$$h_B - h_A = \mathbf{0.15 \text{ m (Ans.)}}$$

$$\text{or, } \frac{P_B}{w} - \frac{P_A}{w} = 0.15 \quad \text{or } P_B - P_A$$

$$= w \times 0.15 = 9.81 \times 0.15 = 1.47 \text{ kN/m}^2 = \mathbf{1.47 \text{ kPa (Ans.)}}$$

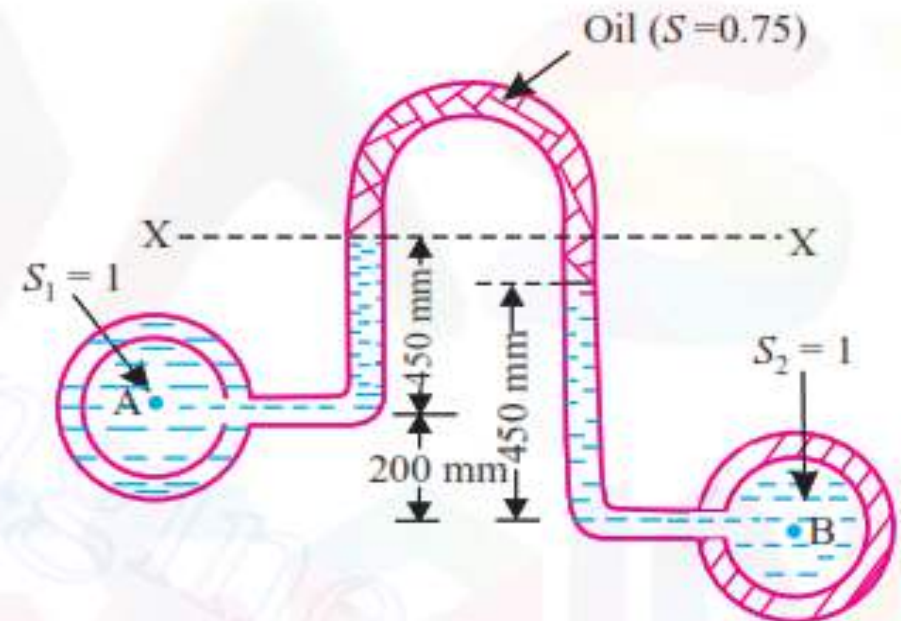


Fig. 2.26

Advantages and Disadvantages of Manometers

Advantages

1. Easy to fabricate and relatively inexpensive
2. Can withstand vibrations
3. Has good accuracy
4. Requires little maintenance
5. High sensitivity
6. Suitable for low pressure and low differential pressures
7. Flexibility to change of sensitivity by changing quantity of manometric liquid in the manometer

Limitations:

1. Usually bulky and large in size
2. Fragile and can break easily
3. Requires accurate measurement of meniscus
4. Surface tension of the manometric fluid can cause capillary effect
5. Changes in temperatures, altitude and gravity can affect the readings of manometers

Mechanical Gauges

The monometers discussed previously are suitable for comparatively low pressures.

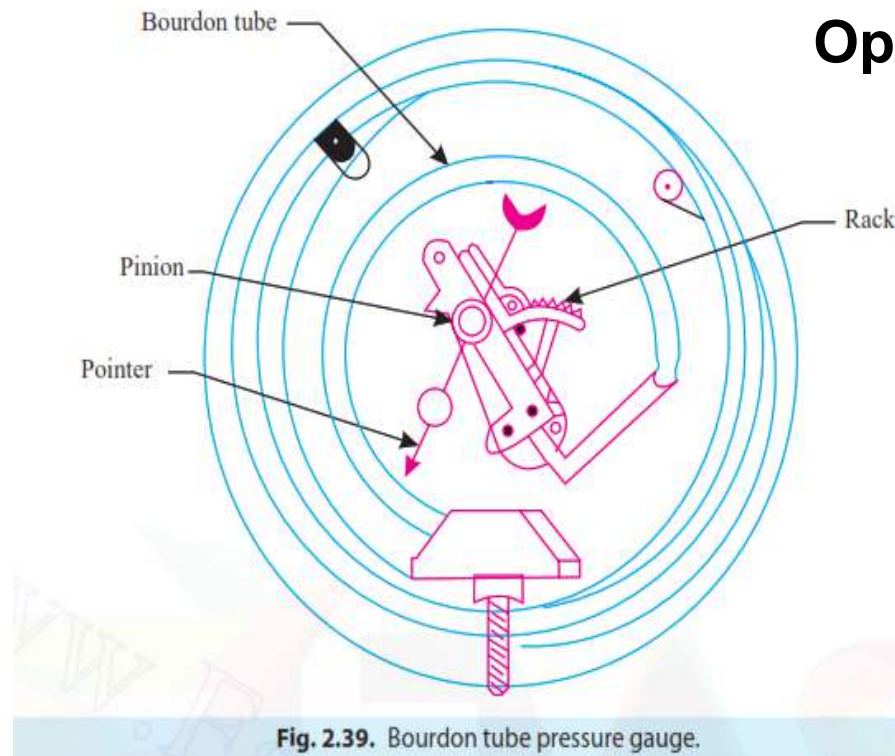
Manometers may become unnecessarily larger to use for high pressures even when filled with heavy liquids. Some important types of these gauges are:

- i. Bourdon tube pressure gauge
- ii. Diaphragm gauge and
- iii. Vacuum gauge.

1. Bourdon Tube Pressure Gauges

- Used for measuring high as well as low pressures
- *A simple form of this kind of gauge is shown in Fig. 2.39*
- *Bourdon tubes are generally made of bronze or nickel steel.*
- *Bronze bourdon tubes are generally used for low pressures while Nickel tubes are used for high pressures.*
- *Based on the purpose of the Bourdon tube gauge, the gauges may take different forms such as:*

- (i) *Compound Bourdon Tube* - used for measuring pressures both above and below atmospheric pressure.
- (ii) *Double Bourdon Tube* – Used where vibrations are encountered.



Operation

- The elliptical metal tube responds to pressure changes
- *The free end of the tube moves and is in turn connected by suitable levers to a rack, which engages with a small pinion mounted on the same spindle as the pointer.*
- *Pressure applied to the tube causes the rack and pinion to move*
- *The pressure is indicated by the pointer over a dial which can be graduated in a suitable scale.*

2. Diaphragm Gauge

- This type of gauge uses metallic disc or diaphragm instead of bent tube
- *The disc or diaphragm is used for actuating the device (See fig 2.40)*

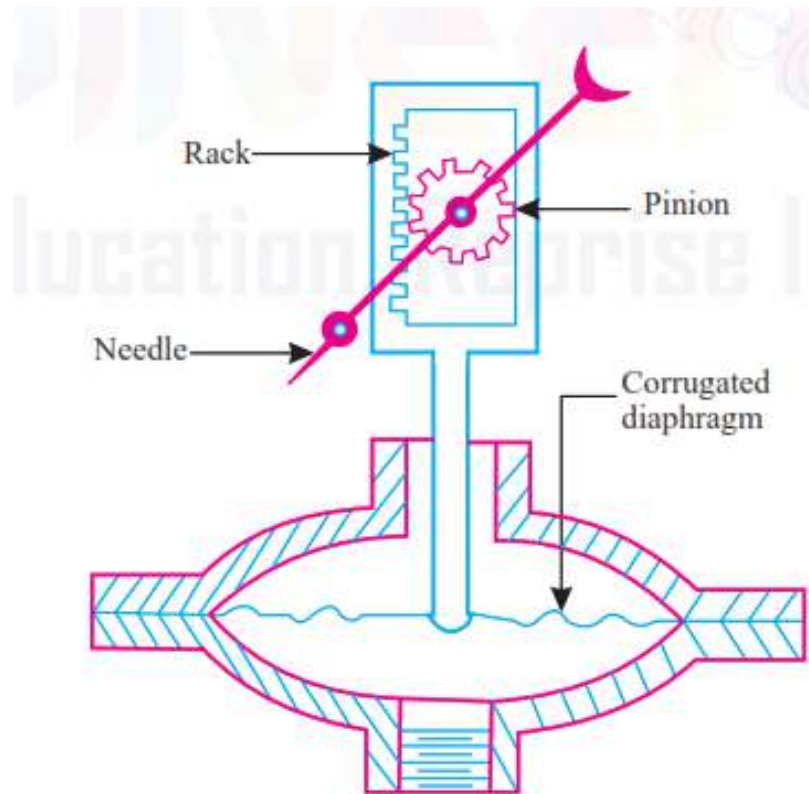


Fig. 2.40. Diaphragm gauge.

Operation

- When pressure is applied on the lower side of the diaphragm it is deflected upward.
- *The movement of diaphragm is transmitted to a rack and pinion*
- *The pinion is attached to the spindle of needle moving on a graduated dial.*
- *The dial can be graduated in to a suitable scale.*

2. Vacuum Gauge

- The bourdon gauges discussed earlier can be used to measure vacuum instead of pressure.
- *Vacuum gauge is a modification of the bourdon gauge.*
- *Thus, in this case, the tube be bent inward instead of outward as in pressure gauges*
- *Vacuum gauges are used to measure the vacuum in the condensers, etc. If there is leakage, the vacuum will drop.*

The pressure gauge installation requires the following considerations:

- Flexible copper tubing and compression fittings are recommended for most installations.
- The installation of a gauge cock and tee in the line close to the gauge is recommended – this permits the gauge to be removed for testing and replacement
- *Pulsating pressures in the gauge line are not required*
- *The gauge and its connecting line is filled with an inert liquid and as such, liquid seals are provided.*

Pressure at a Point in Compressible Fluid

- In compressible fluids, density changes with change of pressure and temperature
- *Such problems are experienced in aeronautics, oceanography and meteorology where we deal with atmospheric air where density, pressure and temperature changes with elevation.*
- *For gases, the equation of state is:*

$$\frac{P}{\rho} = RT$$

$$\rho = \frac{P}{RT}$$

Recall from pressure
var. in fluids

$$\frac{dp}{dZ} = w = \rho g = \frac{P}{RT} \times g$$

$$\frac{dp}{P} = \frac{g}{RT} dZ$$

From the pressure eq.,
pressure is measured
vertically downward, but
where Z is measured
vertically up, then

$$\frac{dp}{dZ} = -\rho g \quad \text{Hence,}$$

$$\frac{dp}{P} = \frac{-g}{RT} dZ$$

In isothermal Process – If temperature T is constant then;

$$\int_{p_0}^p \frac{dp}{p} = - \int_{Z_0}^Z \frac{g}{RT} dz = - \frac{g}{RT} \int_{Z_0}^Z dz$$

$$\log \frac{p}{p_0} = \frac{-g}{RT} [Z - Z_0]$$

where p_0 is the pressure where height is Z_0 . If the datum line is taken at Z_0 , then $Z_0 = 0$ and p_0 becomes the pressure at datum line.

$$\log \frac{p}{p_0} = \frac{-g}{RT} Z$$

$$\frac{p}{p_0} = e^{-gZ/RT}$$

Or pressure at a height Z is given by $p = p_0 e^{-gZ/RT}$

In Adiabatic Process

If temperature T is varying and the process follows adiabatic law then the pressure - density relationship is given by:

$$\frac{p}{\rho^k} = \text{Constant} = C \quad \dots\dots(i) \quad \text{where } k \text{ is a ratio of specific constant}$$

$$\rho^k = \frac{p}{C}$$

$$\rho = \left(\frac{p}{C}\right)^{1/k}$$

From eq. of pressure var. in fluids,

$$\frac{dp}{dZ} = -\rho g = -\left(\frac{p}{C}\right)^{1/k} g$$
$$\frac{dp}{\left(\frac{p}{C}\right)^{1/k}} = -g dZ \text{ or } C^{1/k} \frac{dp}{p^{1/k}} = -g dZ$$

Integrating, we get $\int_{p_0}^p C^{1/k} p^{-1/k} dp = \int_{Z_0}^Z -g dZ$

$$C^{1/k} \left[\frac{p^{-1/k+1}}{-\frac{1}{k}+1} \right]_{p_0}^p = -g [Z]_{Z_0}^Z$$

$$\left[\frac{C^{1/k} p^{-1/k+1}}{-\frac{1}{k}+1} \right]_{p_0}^p = -g [Z]_{Z_0}^Z$$

From eq. (1) $C^{1/k} = \left(\frac{p}{\rho^k} \right)^{1/k} = \frac{p^{1/k}}{\rho}$ Substituting $C^{1/k}$ in eq (2) we get

$$\left[\frac{p^{1/k}}{\rho} \frac{p^{-1/k+1}}{-\frac{1}{k}+1} \right]_{p_0}^p = -g[Z - Z_0]$$

$$\left[\frac{p^{\frac{1-1+k}{k}}}{\rho^{\frac{k-1}{k}}} \right]_{p_0}^p = -g[Z - Z_0] \text{ or } \left[\frac{k}{k-1} \frac{p}{\rho} \right]_{p_0}^p = -g[Z - Z_0]$$

$$\frac{k}{k-1} \left[\frac{p}{\rho} - \frac{p_0}{\rho_0} \right] = -g[Z - Z_0]$$

If datum line is taken at Z_0 , where pressure, temperature and density are p_0 , T_0 and ρ_0 then $Z_0=0$

$$\frac{k}{k-1} \left[\frac{p}{\rho} - \frac{p_0}{\rho_0} \right] = -gZ \text{ or } \frac{p}{\rho} - \frac{p_0}{\rho_0} = -gZ \frac{(k-1)}{k}$$

$$\frac{p}{\rho} = \frac{p_0}{\rho_0} - gZ \frac{(k-1)}{k} = \frac{p_0}{\rho_0} \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]$$

or
$$\frac{p}{\rho} \times \frac{\rho_0}{p_0} = \left[1 + \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]$$

But from equation (i),
$$\frac{p}{\rho^k} = \frac{p_0}{\rho_0^k} \quad \text{or} \quad \left(\frac{\rho_0}{\rho} \right)^k = \frac{p_0}{p} \quad \text{or} \quad \frac{\rho_0}{\rho} = \left(\frac{p_0}{p} \right)^{1/k}$$

Substituting the value of $\frac{\rho_0}{\rho}$ in equation (iii), we get

$$\frac{p}{p_0} \times \left(\frac{p_0}{p} \right)^{1/k} = \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]$$

or
$$\frac{p}{p_0} \times \left(\frac{p}{p_0} \right)^{-1/k} = \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]$$

or
$$\left(\frac{p}{p_0} \right)^{1 - \frac{1}{k}} = \left(\frac{p}{p_0} \right)^{\frac{k-1}{k}} = \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]$$

$\therefore \frac{p}{p_0} = \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]^{\frac{k}{k-1}} \quad \text{.....eq (3)}$

$\therefore$ Pressure at a height Z from ground level is given by

$$p = p_0 \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]^{\frac{k}{k-1}} \quad \begin{array}{l} p_0 = \text{pressure at ground level, where } Z_0 = 0 \\ \rho_0 = \text{density of air at ground level} \end{array}$$

Consider equation of state
$$\frac{p_0}{\rho_0} = RT_0 \quad \text{or} \quad \frac{\rho_0}{p_0} = \frac{1}{RT_0}$$

$$p = p_0 \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0} \right]^{\frac{k}{k-1}}$$

Temperature at a point in compressible fluid

For adiabatic process, the temperature at any height in air is calculated as:

$$\frac{p_0}{\rho_0} = RT_0 \text{ and } \frac{p}{\rho} = RT$$

Dividing these equations, we get

$$\left(\frac{p_0}{\rho_0}\right) \div \frac{p}{\rho} = \frac{RT_0}{RT} = \frac{T_0}{T} \quad \text{or} \quad \frac{p_0}{\rho_0} \times \frac{\rho}{p} = \frac{T_0}{T}$$
$$\frac{T}{T_0} = \frac{\rho_0}{p_0} \times \frac{p}{\rho} = \frac{p}{p_0} \times \frac{\rho_0}{\rho}$$

Recall eq (3)

$$\frac{p}{p_0} = \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{\frac{k}{k-1}}$$

For Adiabatic process

$$\frac{p}{\rho^k} = \frac{p_0}{\rho_0^k} \text{ or } \left(\frac{\rho_0}{\rho}\right)^k = \frac{p_0}{p}$$

$$\frac{\rho_0}{\rho} = \left(\frac{p_0}{p}\right)^{\frac{1}{k}} = \left(\frac{p}{p_0}\right)^{-\frac{1}{k}}$$
$$= \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{\left(\frac{k}{k-1}\right) \times \left(-\frac{1}{k}\right)} = \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{-\frac{1}{k-1}}$$

Substituting the values of $\frac{p}{p_0}$ and $\frac{\rho_0}{\rho}$ in equation

$$\begin{aligned}\frac{T}{T_0} &= \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{\frac{k}{k-1}} \times \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{-\frac{1}{k-1}} \\ &= \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right]^{\frac{k}{k-1} - \frac{1}{k-1}} = \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right] \\ T &= T_0 \left[1 - \frac{k-1}{k} \frac{gZ}{RT_0}\right] \quad \text{.....eq (4)}\end{aligned}$$

Worked Example:

1. The barometric pressure at sea level is 760 mm of mercury while that on a mountain top is 735 mm. If the density of air is assumed constant at 1.2 kg/m^3 , what is the elevation of the mountain top

Solution. Given: Pressure at sea level, $p_0 = 760 \text{ mm of Hg}$

$$= \frac{760}{1000} \times (13.6 \times 1000) \times 9.81 = 101396 \text{ N/m}^2$$

Pressure at mountain, $p = 735 \text{ mm of Hg}$

$$= \frac{735}{1000} \times (13.6 \times 1000) \times 9.81 = 98060 \text{ N/m}^2$$

Density of air, $\rho = 1.2 \text{ kg/m}^3$

Height of the mountain top from sea-level, h :

It is a known fact that as the elevation above the sea-level increases, the atmospheric pressure decreases. As the density is constant (given), hence the pressure at any height ' h ' above the sea-level is given by the equation,

$$p = p_0 - \rho gh \quad \text{or} \quad h = \frac{p_0 - p}{\rho g} = \frac{(101396 - 98060)}{1.2 \times 9.81} = 283.4 \text{ m (Ans.)}$$

Worked Example:

2.If the pressure at sea level is 10.143 N/cm². determine the pressure at a height of 2500 m assuming the pressure variation follows i) Hydrostatic law and ii) isothermal law. Take the density of air as 1.208 kg/m³

Solution. Given :

Pressure at sea-level,

$$p_0 = 10.143 \text{ N/cm}^2 = 10.143 \times 10^4 \text{ N/m}^2$$

Height,

$$Z = 2500 \text{ m}$$

Density of air,

$$\rho_0 = 1.208 \text{ kg/m}^3$$

(i) **Pressure by hydrostatic law.** For hydrostatic law, ρ is assumed constant and hence p is given

by equation $\frac{dp}{dZ} = -\rho g$

Integrating, we get

$$\int_{p_0}^p dp = \int -\rho g dZ = -\rho g \int_{Z_0}^Z dZ$$

or

$$p - p_0 = -\rho g [Z - Z_0]$$

For datum line at sea-level,

$$Z_0 = 0$$

$\therefore$

$$\begin{aligned} p - p_0 &= -\rho g Z \quad \text{or} \quad p = p_0 - \rho g Z \\ &= 10.143 \times 10^4 - 1.208 \times 9.81 \times 2500 \quad [\because \rho = \rho_0 = 1.208] \\ &= 101430 - 29626 = 71804 \frac{\text{N}}{\text{m}^2} \quad \text{or} \quad \frac{71804}{10^4} \text{ N/cm}^2 \\ &= \mathbf{7.18 \text{ N/cm}^2. \text{ Ans.}} \end{aligned}$$

(ii) **Pressure by Isothermal Law.** Pressure at any height Z by isothermal law is given by equation (2.18) as

$$\begin{aligned} p &= p_0 e^{-gZ/RT} \\ &= 10.143 \times 10^4 e^{-\frac{gZ \times \rho_0}{p_0}} \quad \left[\because \frac{p_0}{\rho_0} = RT \text{ and } \rho_0 g = w_0 \right] \\ &= 10.143 \times 10^4 e^{-\frac{Z \rho_0 \times g}{p_0}} \\ &= 10.143 \times 10^4 e^{-(2500 \times 1.208 \times 9.81)/10.143 \times 10^4} \\ &= 101430 \times e^{-.292} = 101430 \times \frac{1}{1.3391} = 75743 \text{ N/m}^2 \\ &= \frac{75743}{10^4} \text{ N/cm}^2 = \mathbf{7.574 \text{ N/cm}^2. \text{ Ans.}} \end{aligned}$$

Worked Example:

3. An aeroplane is flying at an altitude of 500 m. Calculate the pressure around the aeroplane, given the lapse-rate in the atmosphere as 0.0065°K/m . Neglect variation of g with altitude. Take pressure and temperature at ground level as 10.143 N/cm^2 and 15°C and density of air as 1.285 kg/cm^3 .

Solution. Given :

Height, $Z = 5000 \text{ m}$

Lapse-rate, $L = \frac{dT}{dZ} = - .0065^\circ\text{K/m}$

Pressure at ground level, $p_0 = 10.143 \times 10^4 \text{ N/m}^2$
 $t_0 = 15^\circ\text{C}$

$\therefore T_0 = 273 + 15 = 288^\circ\text{K}$

Density, $\rho_0 = 1.285 \text{ kg/m}^3$

$$\begin{aligned}\therefore \text{Temperature at 5000 m height} &= T_0 + \frac{dT}{dZ} \times \text{Height} = 288 - .0065 \times 5000 \\ &= 288 - 32.5 = 255.5^\circ\text{K}\end{aligned}$$

First find the value of power index k as

$$\text{From equation (2.22), we have } L = \frac{dT}{dZ} = -\frac{g}{R} \left(\frac{k-1}{k} \right)$$

$$\text{or } - .0065 = -\frac{9.81}{R} \left(\frac{k-1}{k} \right)$$

$$\text{where } R = \frac{p_0}{\rho_0 T_0} = \frac{101430}{1.285 \times 288} = 274.09$$

$$\therefore - .0065 = -\frac{9.81}{274.09} \left(\frac{k-1}{k} \right)$$

$$\therefore k = 1.222$$

The pressure is given by equation (2.19) as

$$\begin{aligned} p &= p_0 \left[1 - \frac{k-1}{k} gZ \frac{\rho_0}{p_0} \right]^{\left(\frac{k}{k-1} \right)} \\ &= 101430 \left[1 - \left(\frac{1.222 - 1.0}{1.222} \right) \times 9.81 \times \frac{5000 \times 1.285}{101430} \right]^{\frac{1.222}{1.222 - 1.0}} \\ &= 101430 \left[1 - \frac{.222}{1.222} \times 9.81 \times \frac{5000 \times 1.285}{101430} \right]^{\frac{1.222}{.222}} \\ &= 101430 [1 - 0.11288]^{5.50} = 101430 \times 0.5175 = 52490 \text{ N/m}^3 \\ &= \mathbf{5.249 \text{ N/cm}^2}. \text{ Ans.} \end{aligned}$$